

Introduction to Urban Economics

EC 3335, Set 01

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Prologue

Schedule

Today

- Welcome, check in, survey, and syllabus
- Introduction: *What is (urban) economics?*
- Toolkit: *What is a model?*

Upcoming

- EC2301 review
- Principals of urban economics

Introduction: About me

Name: Chen Xu. Feel free to call me **Chen**

- Office: TRUS 309, Office Hours: **TBD**, Email: cxu@stedwards.edu

Area of Interests:

- **Research:**
 - Applied micro topics related to labor economics
 - financial Economics, casual inference
- **Teaching:**
 - Economics, Statistics, Data analysis

Introduction: About me (cont.)

Not school:

- Grew up in China
- I like basketball 🏀, recently fall in love with fishing 🐟👍

Introduction: About you

I hope that you:

- (i) Are an eager student ready to learn about urban economics
- (ii) Have passed EC2301 with at least **some recollection** of the material
- (iii) Engage in class (ask lots of questions) and support your peers
- (iv) Will read my **syllabus** (especially before sending me an email)

I want to discuss the following:

- (i). Course policies
- (ii). Grades and grading

Syllabus

Syllabus: Attendance

Attendance is "**mandatory**". However,

- everyone has up to 3 absences
- your attendance grade will be **deteriorated** after you miss more than 3 classes
- watch **recorded** lectures to catch up

Attendance is an efficient use of your time

- Mixture of **slide decks** and **in-class examples** on the board

Exams will be administered during class time

Syllabus: Grading

Your grade will be determined by:

- **5%:** Attendance
- **35%:** Problem sets (5 assignments)
- **20%:** Midterm Exam
- **20%:** Lab Quiz
- **20%:** Final Group Project[†]

Q: Is there a curve?

A: Typically yes. At the end of the term the class will be curved

[†] More details coming soon

Syllabus: Grading

Caveat: Due to a curve, standing in this course will depend on your score relative to your peers--I may not be able to give you an accurate assessment of your grade

From the syllabus:

Your grade will be determined relative to your peers, so during the course, I will not be able to tell you what your exact letter grade is at any point in time, because it depends on everyone's overall scores of the class.

Syllabus: Exams

Midterm: TBD (in class)

Absolutely no makeups

- **Unless** you have a valid excuse
- Must contact me **before** the exam with supporting documentation

There will be **4** problem sets throughout the term:

- Signal what to expect on the exams
- You must submit a **.pdf** document on Canvas
- The lowest scored problem set is dropped
- No late homework assignments will be accepted
- If your scan is bad, or your handwriting illegible, you get **nothing**

Syllabus: Canvas + course site

I use Canvas in conjunction to a simple course website hosted on GitHub

Canvas:

- **All** assignments submissions and announcements

Course site:

- Light, public page with links to all course material
- Do not need an account
- I use it out of convenience and other personal reasons
- **ONLY** thing on the page that will not be on Canvas is the slide decks

What is economics?

What is economics?

The Wealth of Nations by Adam Smith (1776)

Seminal work on economics, four volumes; comparable to the works of:

- Newton's *Principia Mathematica*
- Darwin's *Origin of Species*

Wealth ("weal")

- Money and other assets
- But also well-being, welfare

Archaic use of "wealth"

- Economics is not merely the study of how to get rich
- Please don't ask me about the stock market

What is economics?

In a nutshell:

Economics is the study of how people make allocation decisions to maximize their happiness when facing limited resources (budgets)

There is never **enough resources**; we call this **scarcity**

Scarcity gives us constraints to which we do the best *we can*

Other examples of scarcity

- Time
- Health
- Land
- Natural resources

Economics is very general; the market lens can be used across many topics

Intro to Urban Economics

Intro to Urban Economics

A mash-up between **economics** and **geography**

Economics: Study of how people and firms allocate scarce resources.

- Individual utility, markets, trade, welfare

Geography: Studies effects of location and the environment

- Hydrology, climate, resources, etc.

Economics + Geography : Study of how individuals and firms choose utility and profit maximizing locations, and consequences of these decisions

Intro to Urban Economics

We will study how the **distribution** of people & firms across space impacts:

- City growth 
- Crime 
- The environment 
- Income growth & inequality 
- Education 
- Employment + wages 

We will also examine the efficacy of various **place - based policies**

- Minimum Wage
- Rent Control
- etc...
- Land Use Restrictions
- Sustainability

What are the economic drivers behind urban development?

This Course

This class has two fairly distinct halves:

1. **Philosophy & Tools**

- Why do cities exist? Why do they grow? Why do they decline?
- Fundamental tools of labor & urban econ (supply and demand)

2. **Application**

- Rent Control & Minimum Wage
- Highways and urban transportation
- Income inequality and environmental issues

What is a city?

According to the Census Bureau...

Urban Area: a densely settled geographical area with:

- Minimum population of **2,500**
- Minimum density of **500 people per square mile**

Metropolitan Area: an urbanized area with at - least **50k population**

Micropolitan Area: an urbanized area with at least **10k** but not as many as **50k** people

MSA: abbrev. for both **metropolitan** and **micropolitan** statistical area

Principal City: the **largest municipality in an MSA**

City: Dense collection of people in specific geographic area

Intro to Urban: Cities

The majority of the US population lives in cities[†] and more people will continue to migrate into urban areas

Questions:

1. Do you like cities?
2. What are favorite city **amenities**?
3. Are cities at odds with the natural world?

Why do cities exist?

Location matters

[†]. 80 - ish percent, according to the Census Bureau

Intro to Urban: Location...

Where you live has implications for

- Your contribution to **global carbon emissions**
 - *Why does this vary across cities?*
- Your **wage** and **rent**
 - *Why does this vary across cities?*
- Your **economic mobility**
 - *Why does this vary across cities?*

We will answer these questions as we progress through this course

What is a model?

In this class we will make use of various **mathematical models**

What is a **mathematical model**?

- A model is a description of a system using **math**
- Useful to help **explain** and **predict** behavior

The Canonical Example

$$\text{Supply : } P(Q_s) = 10 + 5 * Q_s$$

$$\text{Demand : } P(Q_d) = 20 - 2 * Q_d$$

This model allows us to make predictions about prices and quantities (from the supply & demand side), *and* the **equilibrium** price and quantity

What is a model?

What are the **pros** of models?

- Allows for us to be very precise with our language
- Gives us the ability to **predict** the various aspects of the economy
- Can shed insight on **mechanisms** through which processes interact

What are the **cons** of models?

- They require assumptions
- **Almost** all assumptions are wrong

But not all wrong assumptions are useless

The ability of the model to **predict data** and **understand mechanisms** determines how useful it is

What is a model?

Did we make assumptions our supply/demand model? **Discuss**

$$\text{Supply : } P(Q_s) = 10 + 5 * Q_s$$

$$\text{Demand : } P(Q_d) = 20 - 2 * Q_d$$

1. **Marginal values** are diminishing and **marginal costs** are increasing [†]

- Generates downward demand and upward supply

1. Demand and Supply are **linear**

2. Demand and Supply are **deterministic**

Are these reasonable? Can your behavior be explained by a simple **linear function**?

Does a function even exist?

†: Marginal = Adding one additional unit

Planning

Next Class:

- EC201 Review
- 5 Axioms of Urban Economics

Reading:

- Get the book ASAP!!
- Read the introduction

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Intro to Urban Economics

1. What is Urban Economics?
2. What is a city?
3. What is a model and why are they useful?